

Aim: Understanding Classes and Object in Java.**1. Write a java program for swapping of two numbers where swap function lies within a class.****CODE:**

```
import java.util.Scanner;
class Swapping{
    int x,y,t;
    Swapping(int a,int b){
        x=a;
        y=b;
        t=x;
        x=y;
        y=t;
    }
    void display(){
        System.out.println("After swapping...");
        System.out.println("x:"+x+", y:"+y);
    }
}
public class Swap{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int a,b;
        System.out.print("Enter value of x:");
        a = sc.nextInt();
        System.out.print("Enter value of y:");
        b = sc.nextInt();
        System.out.println("Before Swapping...");
        System.out.println("x:"+a+", y:"+b);
        Swapping s = new Swapping(a, b);
        s.display();
    }
}
```

OUTPUT:

```
Enter value of x:20
Enter value of y:50
Before Swapping...
x:20, y:50
After swapping...
x:50, y:20
```

2. Write a java program to sort a given array.**CODE:**

```
import java.util.Scanner;
class Sort{
    Scanner s = new Scanner(System.in);
    int n;
    int arr[];
    Sort(){
        System.out.print("Enter the number of element you want to enter:");
        n = s.nextInt();
        arr = new int[n];
    }
}
```

```
void sortArray(){
    System.out.print("Enter the elements:");
    for(int i=0;i<n;i++){
        arr[i]=s.nextInt();
    }
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            if(arr[j]>arr[j+1]){
                int t = arr[j];
                arr[j] = arr[j+1];
                arr[j+1]= t;
            }
        }
    }
}

void display(){
    System.out.println("Your sorted array is...");
    for(int i=0;i<n;i++){
        System.out.print(arr[i]+" ");
    }
}

public class Sorting {
    public static void main(String args[]){
        Sort ob = new Sort();
        ob.sortArray();
        ob.display();
    }
}
```

OUTPUT:

Enter the number of element you want to enter:5

Enter the elements: 9 4 7 1 3

Your sorted array is...

1 3 4 7 9

3. Write a java program to input telephone number and number of calls. Calculate and display the bill amount which include a fixed rent of Rs. 400. The first 150 calls are free with excess calls are charged 80 paise each (0.8 Rupees).

CODE:

```
import java.util.Scanner;
class BillPayer{
    Scanner s = new Scanner(System.in);
    int calls;
    String tel;
    double charge = 400;
    void calculateBill(){
        System.out.print("Enter the telephone number:");
        tel = s.nextLine();
        System.out.print("Enter the number of calls you have done:");
        calls = s.nextInt();
        charge = 400;
        if(calls<=150){
            charge = 400;
        }
    }
}
```

```
    else{
        calls = calls - 150;
        charge = 400 + (calls*0.8);
    }
}
void displayBill(){
    System.out.println("Your bill amount for "+tel+" is "+ charge+" rupees.");
}
}
public class Bill{
    public static void main(String args[]){
        BillPayer b = new BillPayer();
        b.calculateBill();
        b.displayBill();
    }
}
```

OUTPUT:

Enter the telephone number:9869548169

Enter the number of calls you have done:2600

Your bill amount for 9869548169 is 2360.0 rupees.

4. Write a java program to keep a record of height and weight of 10 different people, and find out the number of people whose weight is less than 50 and height is more than 170.

CODE:

```
import java.util.Scanner;
class HandW{
    Scanner s = new Scanner(System.in);
    int h[] = new int[10];
    int w[] = new int[10];
    int n = 10;
    int count = 0;
    void get(){
        System.out.print("Enter 10 heights in cm:");
        for(int i=0;i<n;i++){
            h[i] = s.nextInt();
        }
        System.out.print("Enter 10 weights in kg:");
        for(int i=0;i<n;i++){
            w[i] = s.nextInt();
        }
    }
    void display(){
        System.out.print("Person whose weight is less than 50kg and height greater than 150cm are ");
        for(int i=0;i<n;i++){
            if(h[i]>=150 && w[i]<50){
                count = 1 + count;
            }
        }
        System.out.print(count);
    }
}
public class HeightWeigth {
    public static void main(String args[]){
        HandW hw = new HandW();
```

```
hw.get();  
hw.display();  
}  
}
```

OUTPUT:

Enter 10 heights in cm:160 140 155 170 149 180 130 152 165 159

Enter 10 weights in kg:45 60 49 55 48 35 52 47 44 50

Person whose weight is less than 50kg and height greater than 150cm are 5

5. Write a java program containing following classes: Class Marks containing marks in 2 papers of a subject. Use parameterized constructor to initialize the values. Class Marks1 defines an object of the above class and pass the parameters of your choice. Define an instance variable name of the student. Write a method to find the average of 2 papers and a method to display the name of the student marks in 2 papers and average marks of the student. Class MarksDemo contains main() method to invoke the members of class Marks1.

CODE:

```
class Marks{  
    int m1,m2;  
    Marks(int x,int y){  
        m1 = x;  
        m2 = y;  
    }  
}  
  
class Marks1{  
    Marks mk;  
    String name;  
    Marks1(String nm,int m1,int m2){  
        name = nm;  
        mk = new Marks(m1,m2);  
    }  
    double avg(){  
        return (mk.m1+mk.m2)/2;  
    }  
    void display(){  
        System.out.println("Name:"+name);  
        System.out.println("Average:"+avg());  
        System.out.println("Marks in Subject1:"+mk.m1);  
        System.out.println("Marks in Subject2:"+mk.m2);  
    }  
}  
  
public class MarksDemo {  
    public static void main(String args[]){  
        Marks1 m = new Marks1("Atomic",90,80);  
        m.display();  
    }  
}
```

OUTPUT:

Name:Atomic

Average:85.0

Marks in Subject1:90

Marks in Subject2:80